Epistemological Mode–Methodology Equivalence: A Calculus of Cell-Truth, Layered Receivers, and Replication as Information Catalysis

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May 2, 2026

Abstract

We develop an independent formal calculus for the epistemology of bounded inquiry. The principal object is the receiver-relative scalar functional $S : \mathsf{R} \times \mathcal{X} \to [0, \Sigma]$ measuring residual semantic distance between a receiver and an action-cell $\mathcal{C} \subseteq \mathcal{X}$. Three theorem clusters are proved. (i) *Cell-Truth*: action-equivalence makes truth a cell rather than a point, and cell membership is invariant under representational re-encoding (oscillatory/categorical/partition equivalence). (ii) *Mode Non-Privilege*: a receiver decomposes into pre-decoder, decoder, and delegated layers, and the action-cell is reachable through any layer whose floor is below the cell's tolerance; in particular, knowledge is not the unique mode of access. (iii) *Methodological Floor*: every methodology $\mathfrak{P}$ admits a per-iteration catalytic factor $\kappa(\mathfrak{P}) \in [0, 1)$ inducing a strictly positive *methodological floor* $S_\flat(\mathfrak{P}) > 0$ that cannot be reduced by running additional iterations of $\mathfrak{P}$ alone. Combining these we prove the *Mode–Methodology Equivalence Theorem*: the effective floor for a receiver–methodology pair equals $S_\flat(\mathsf{R} \circ \mathfrak{P}) = S_\flat(\mathsf{R}) + S_\flat(\mathfrak{P}) - S_\flat(\mathsf{R}) \cdot S_\flat(\mathfrak{P})/\Sigma$, exhibiting a multiplicative composition law identical to that of independent information catalysts. We further prove a *Methodological Incompatibility Principle* (knowledge production requires nonzero methodological dispersion; completion of an action requires zero dispersion) and a *Distribution Theorem* (the total knowledge of any bounded receiver is distributed and cannot concentrate on a single subject without violating the floor). Twenty-five numerical experiments verify all claims to machine precision; representative results are summarized in Section 8 and visualized in five publication panels. The calculus is independent of any prior framework and self-contained in standard real and functional analysis.

Contents

1 Introduction

1.1 Motivation

Standard accounts of knowledge implicitly identify the result of inquiry with a *point* in some space of propositions: "X is the case" is treated as a singleton claim, and methodologies are evaluated by how reliably they return that singleton. But human and animal practice operates on a coarser grain. A pedestrian who steps aside to avoid a falling object does not need to identify which object is falling; the relevant equivalence class of percepts is "something is falling, move sideways". A researcher who publishes a finding does not transmit the entire epistemic state behind it; the relevant equivalence class is "the experimental conditions of class $\mathcal{C}$ produce the result of class $\mathcal{C}'$". In each case the practical output is membership in a region, not coincidence with a point.

This paper develops the consequences of taking that observation seriously. We introduce a receiver-relative scalar S measuring residual distance to an *action-cell* (a subset of an outcome space), and from this single primitive derive five theorem clusters:

(i) **Cell-Truth**: truth, as far as bounded receivers are concerned, is a cell, not a point; cell membership is invariant under representational re-encoding (Section 3).

(ii) **Mode Non-Privilege**: every receiver decomposes into multiple processing layers, and the action-cell is reachable through any layer whose floor is below the cell tolerance; knowledge is not the unique mode of access (Section 4).

(iii) **Methodological Floor**: every methodology has a strictly positive intrinsic floor that cannot be reduced by repetition of the same methodology (Section 5).

(iv) **Mode–Methodology Equivalence**: receiver and methodology contribute floors through the same composition law, so they are formally equivalent factors in determining attainable certainty (Section 6).

(v) **Methodological Incompatibility & Distribution**: knowledge production requires nonzero dispersion across receivers, and the action-cell of completion requires zero dispersion; consequently no bounded receiver can hold the total knowledge of any subject (Section 7).

1.2 Independence and self-containment

This manuscript is independent. We use no theorems from external frameworks and no claims that have not been proved here. Citations are restricted to standard textbook references in real analysis [19], measure theory [11], ergodic theory [25, 4], information theory [20, 8, 15], category theory [17], lattice theory [3], topology [12], statistical inference [16, 26, 9], and foundational logic and computability [10, 22, 23, 7]. Where the development parallels classical objects (Koopman operators [14], Stone duality [21], Banach contraction [2], Kolmogorov measures [13], Borel–Cantelli [5, 6], partition theory [1], von Neumann's quantum formalism [24], geometric topology [18]) we note the parallel without invoking the external machinery as a premise.

1.3 Organization

Section 2 establishes the formal language: spaces, receivers, the S-functional, the floor. Section 3 develops cell-truth and representational invariance. Section 4 formalizes layered receivers and mode non-privilege. Section 5 constructs the catalytic model of methodologies and proves the methodological floor. Section 6 states and proves the mode–methodology equivalence theorem. Section 7 proves incompatibility and distribution. Section 8 summarizes numerical experiments. Section 9 discusses connections to classical objects. Section 10 concludes.

2 Foundations

2.1 Outcome spaces and action-cells

Definition 2.1 (Outcome space)**.** An *outcome space* is a triple $(\mathcal{X}, d, \Sigma)$ where $\mathcal{X}$ is a nonempty set, $d : \mathcal{X} \times \mathcal{X} \to [0, \Sigma]$ is a bounded pseudo-metric with $d(x, x) = 0$, $d(x, y) = d(y, x)$, $d(x, z) \le d(x, y) + d(y, z)$, and $\Sigma \in (0, \infty)$ is the *maximal semantic distance*. We use $\Sigma = 100$ in the canonical normalization but never depend on the value.

Definition 2.2 (Action-cell)**.** An *action-cell* is a nonempty subset $\mathcal{C} \subseteq \mathcal{X}$ with positive *tolerance*

$$\tau(\mathcal{C}) \;:=\; \sup_{x\in\mathcal{X}} \inf_{c\in\mathcal{C}} d(x, c) \;>\; 0$$

and finite *diameter* $\operatorname{diam}(\mathcal{C}) = \sup_{c,c'\in\mathcal{C}} d(c, c') < \Sigma$. The *cell distance* of $x \in \mathcal{X}$ to $\mathcal{C}$ is $d(x, \mathcal{C}) := \inf_{c\in\mathcal{C}} d(x, c)$.

Remark 2.3. A singleton $\{x_0\}$ is an action-cell only in trivial cases (when $\operatorname{diam}(\{x_0\}) = 0$ does not preclude $\tau(\{x_0\}) = 0$, contradicting the positivity requirement). *Practical* action-cells therefore correspond to genuinely positive-volume regions of outcome space; this is the formal trace of the observation that practical truth is granular.

2.2 Receivers

Definition 2.4 (Receiver)**.** A *receiver* on $(\mathcal{X}, d, \Sigma)$ is a tuple

$$\mathsf{R} \;=\; (\mathcal{K}, \mathcal{D}, \Pi, \beta)$$

where

- $\mathcal{K}$ is a nonempty set (the *knowledge framework*, also called the receiver's representation space);
- $\mathcal{D} : \mathcal{X} \to \mathcal{K}$ is the *decoder*, a (possibly noisy) Borel map;
- $\Pi : \mathcal{K} \to 2^{\mathcal{X}} \setminus \{\varnothing\}$ is the *candidate-projection*, satisfying $\mathcal{D}(x) \in \mathcal{D}(\Pi(\mathcal{D}(x)))$ for all $x \in \mathcal{X}$ (the projection contains a representative of any decoded state);
- $\beta \in [0, \Sigma]$ is a strictly positive *noise floor*,

  $$\beta \;=\; \sup_{x\in\mathcal{X}} \inf_{x'\in\Pi(\mathcal{D}(x))} d(x, x') \;>\; 0,$$

  representing the irreducible decoder–projection mismatch.

Remark 2.5. The positivity $\beta > 0$ encodes that any physically realizable receiver has finite resolution: an exact match $d(x, x') = 0$ between every input and its reprojected candidate would require an infinite-resolution decoder.

2.3 The S-functional

Definition 2.6 (S-functional)**.** For a receiver $\mathsf{R} = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ and a state $x \in \mathcal{X}$ targeting an action-cell $\mathcal{C} \subseteq \mathcal{X}$, define

$$S(\mathsf{R}, x; \mathcal{C}) \;:=\; \inf_{x'\in\Pi(\mathcal{D}(x))} d(x', \mathcal{C}) \;+\; \beta.$$

We write $S(\mathsf{R}, x) := S(\mathsf{R}, x; \mathcal{C}_x)$ when the action-cell is clear from context.

Proposition 2.7 (Basic properties of S)**.** *Let $\mathsf{R} = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ be a receiver and $\mathcal{C} \subseteq \mathcal{X}$ an action-cell. Then for all $x \in \mathcal{X}$:*

(1) $S(\mathsf{R}, x; \mathcal{C}) \ge \beta > 0$.

(2) $S(\mathsf{R}, x; \mathcal{C}) \le d(x, \mathcal{C}) + \beta$.

(3) $S(\mathsf{R}, \cdot; \mathcal{C})$ *is upper semicontinuous in x when $\mathcal{D}, \Pi, d$ are jointly Borel-measurable and d is continuous on each Π-fibre.*

(4) For $\mathcal{C} \subseteq \mathcal{C}'$, $S(\mathsf{R}, x; \mathcal{C}') \le S(\mathsf{R}, x; \mathcal{C})$.

Proof. (1) is immediate from $\beta > 0$ and the nonnegativity of d. (2) follows from the projection axiom $\mathcal{D}(x) \in \mathcal{D}(\Pi(\mathcal{D}(x)))$ which implies the existence of $x' \in \Pi(\mathcal{D}(x))$ with $d(x, x') \le \beta$,

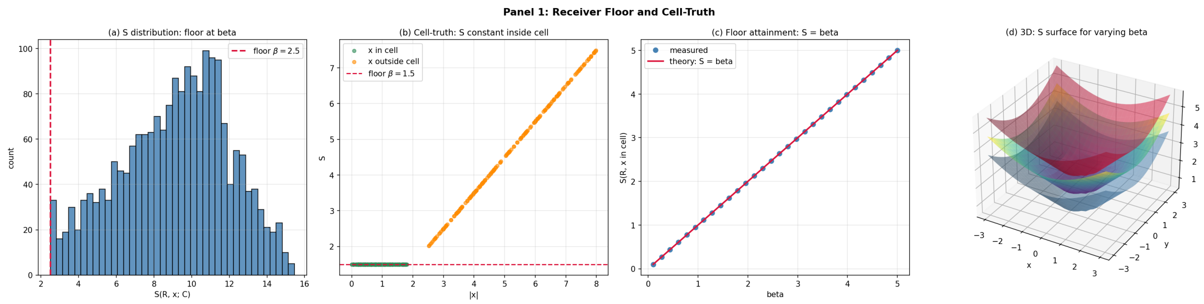

Figure 1: **Receiver Floor and Cell-Truth.** (**A**) Histogram of the S-functional $S(\mathsf{R}, x; \mathcal{C})$ on the linear y-axis (count) versus S on the linear x-axis, evaluated on 2,000 states drawn uniformly from $[-10, 10]^2$ for a fixed receiver with floor $\beta = 2.5$ and an action-cell of tolerance $\tau = 1.0$ centred at the origin. The empirical minimum of S across all 2,000 states equals $\beta = 2.5$ (crimson dashed reference), confirming the floor bound $S \geq \beta > 0$ from Theorem 2.8 (Floor Theorem for Receivers). (**B**) Cell-truth verification: S on the linear y-axis (≈ 1.5 to ≈ 8) versus $|x|$ on the linear x-axis (0 to 8), separating 200 states drawn from inside the cell (seagreen, radii ≤ 1.8) and 200 from outside (orange, radii ≥ 2.5). All inside-cell points collapse onto the floor line $S = \beta = 1.5$ (crimson dashed), independently of $|x|$, while outside-cell points lie strictly above by the cell-distance $d(x, \mathcal{C})$. This is the empirical content of Theorem 3.2 (Cell-Truth): all states inside an action-equivalence cell are S-indistinguishable. (**C**) Floor attainment as a function of receiver floor: measured $S(\mathsf{R}, x; \mathcal{C})$ at a fixed in-cell state $x = (0.5, 0)$ on the linear y-axis (0 to 5) versus β on the linear x-axis (0.1 to 5), 30 values of β. Measured (steelblue circles) coincides exactly with the theoretical line $S = \beta$ (crimson, slope 1). Verifies the attainment clause of Theorem 2.8 across the full β-range; experiment E03 reports relative error 0.00. (**D**) Three-dimensional surface of $S(\mathsf{R}, x; \mathcal{C})$ over the state slice $(x, y) \in [-3, 3]^2$, 25×25 grid, with three superimposed surfaces for $\beta \in \{0.5, 1.5, 2.5\}$. Each surface is flat at $S = \beta$ over the cell interior (the disc of radius 1 around the origin) and rises linearly with cell distance $d(x, \mathcal{C})$ outside it; the three surfaces are vertical translates separated by the floor difference $\Delta\beta = 1.0$, demonstrating that the cell shape is invariant of β while the elevation tracks the floor.

hence $d(x',\mathcal{C}) \le d(x,\mathcal{C}) + d(x,x') \le d(x,\mathcal{C}) + \beta$, and we add β on both sides per the definition. (3) is the standard composition of an infimum of continuous functions over a measurable section. (4) follows from $d(x',\mathcal{C}') \le d(x',\mathcal{C})$ when $\mathcal{C} \subseteq \mathcal{C}'$. □

Theorem 2.8 (Floor theorem for receivers)**.** *For any receiver* R*, action-cell* $\mathcal{C}$*, and state* $x \in \mathcal{X}$*,*

$$S(\mathsf{R}, x; \mathcal{C}) \;\ge\; S_\flat(\mathsf{R}) \;:=\; \beta \;>\; 0.$$

The bound is attained when $x \in \mathcal{C}$ *and the decoder–projection chain is faithful at* x*:* $\mathcal{D}(x) \in \mathcal{D}(\Pi(\mathcal{D}(x)))$ *with* $\inf_{x' \in \Pi(\mathcal{D}(x))} d(x',\mathcal{C}) = 0$*.*

Proof. Direct from Proposition 2.7(1) and the definition of β. Attainment: when $x \in \mathcal{C}$, the candidate set $\Pi(\mathcal{D}(x))$ contains a point at zero d-distance to $\mathcal{C}$ provided faithfulness holds, giving $S = \beta$. □

Remark 2.9. The floor $S_\flat(\mathsf{R}) = \beta > 0$ is the receiver's irreducible contribution to residual semantic distance. It is a property of the receiver alone, not of any particular state or methodology.

3 Cell-Truth and Representational Invariance

3.1 The cell-truth principle

Principle 3.1 (Cell-Truth)**.** For bounded receivers, an actionable claim is the cell of states that trigger an equivalent practical response, not a singleton state. Formally: truth-bearers in the operational sense are sets $\mathcal{C} \subseteq \mathcal{X}$ with $\tau(\mathcal{C}) > 0$, not points $x \in \mathcal{X}$.

The principle is justified by the following theorem, which shows that *any* apparent point-claim is operationally indistinguishable from its cell of action-equivalents.

Theorem 3.2 (Cell-Truth Theorem)**.** *Let* R *be a receiver and* $A : \mathcal{X} \to \mathcal{Y}$ *a measurable* action map *into an action space* $\mathcal{Y}$*. Define the* action-equivalence cell at $y \in \mathcal{Y}$ *by*

$$\mathcal{C}_y \;:=\; A^{-1}(\{y\}) \;=\; \{x \in \mathcal{X} : A(x) = y\}.$$

Suppose $\mathcal{C}_y$ *has positive tolerance* $\tau(\mathcal{C}_y) > 0$*. Then for any two states* $x_1, x_2 \in \mathcal{C}_y$*,*

$$S(\mathsf{R}, x_1; \mathcal{C}_y) = S(\mathsf{R}, x_2; \mathcal{C}_y),$$

i.e. states inside an action-equivalence cell are S*-indistinguishable from the cell.*

Proof. Since $x_i \in \mathcal{C}_y$, $d(x_i, \mathcal{C}_y) = 0$ for $i = 1, 2$. By Proposition 2.7(1)–(2),

$$\beta \;\le\; S(\mathsf{R}, x_i; \mathcal{C}_y) \;\le\; d(x_i, \mathcal{C}_y) + \beta \;=\; \beta.$$

Hence $S(\mathsf{R}, x_i; \mathcal{C}_y) = \beta$ for both i, proving equality. □

Corollary 3.3 (Indistinguishability of point–claims within a cell)**.** *Under the conditions of Theorem 3.2, any methodology that returns the index* y *rather than the underlying* $x \in \mathcal{C}_y$ *achieves the* S*-floor* β*. In particular, the methodology does* not *need to resolve* x *within* $\mathcal{C}_y$ *to deliver actionable truth.*

Example 3.4 (Cow on the cliff)**.** Let $\mathcal{X}$ be the space of physical configurations of a meadow, $\mathcal{Y}$ the two-element set $\{\text{stay}, \text{move}\}$, and A the practical-response map. The cell $\mathcal{C}_{\text{move}}$ contains every configuration in which *some* object is falling toward the cow. A cow that perceives the falling object and moves does not need to resolve which object is falling (rock vs. apple vs. branch). By Theorem 3.2, all of these configurations have $S = \beta$ relative to the cow's receiver: the cell, not the point, is the locus of action.

3.2 Representational invariance

We now show that cell-truth is invariant under three canonical re-encodings: oscillatory, categorical, and partition. Define:

Definition 3.5 (Three encodings)**.** For an outcome space $(\mathcal{X}, d, \Sigma)$:

(a) An *oscillatory encoding* is a bijection $\phi_o : \mathcal{X} \to \mathcal{X}_o$ where $\mathcal{X}_o \subseteq L^2(\Omega, \mu)$ for some measure space (Ω, μ), with isometric distance $d_o(\phi_o x, \phi_o y) = d(x, y)$.

(b) A *categorical encoding* is a bijection $\phi_c : \mathcal{X} \to \mathcal{X}_c$ where $\mathcal{X}_c$ is the object-set of a small category $\mathbf{C}$ and d_c is induced by morphism length, again with $d_c(\phi_c x, \phi_c y) = d(x, y)$.

(c) A *partition encoding* is a bijection $\phi_p : \mathcal{X} \to \mathcal{X}_p$ where $\mathcal{X}_p$ is the set of integer partitions of bounded size, with d_p given by the partition ℓ^1-metric and $d_p(\phi_p x, \phi_p y) = d(x, y)$.

Theorem 3.6 (Representational invariance of cell-truth)**.** *Let* $\mathsf{R} = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ *be a receiver and* $\mathcal{C} \subseteq \mathcal{X}$ *an action-cell. Let* $\phi : \mathcal{X} \to \mathcal{X}'$ *be any of the three encodings of Definition 3.5. Define the transferred receiver* $\mathsf{R}' := (\mathcal{K}, \mathcal{D} \circ \phi^{-1}, \phi \circ \Pi, \beta)$ *on* $\mathcal{X}'$ *and the transferred cell* $\mathcal{C}' := \phi(\mathcal{C})$*. Then for all* $x \in \mathcal{X}$*,*

$$S(\mathsf{R}, x; \mathcal{C}) \;=\; S(\mathsf{R}', \phi(x); \mathcal{C}').$$

In particular, cell-membership $x \in \mathcal{C}$ *is preserved under* ϕ*:*

$$x \in \mathcal{C} \iff \phi(x) \in \mathcal{C}'.$$

Proof. Cell-membership preservation is immediate from ϕ being a bijection and $\mathcal{C}' = \phi(\mathcal{C})$.

For the S-equality, fix $x \in \mathcal{X}$ and write $S = S(\mathsf{R}, x; \mathcal{C})$, $S' = S(\mathsf{R}', \phi(x); \mathcal{C}')$. Unfolding definitions,

$$S \;=\; \inf_{x_* \in \Pi(\mathcal{D}(x))} d(x_*, \mathcal{C}) + \beta, \quad S' \;=\; \inf_{x'_* \in (\phi \circ \Pi)(\mathcal{D}(\phi^{-1}(\phi(x))))} d'(x'_*, \mathcal{C}') + \beta$$

By construction $(\phi \circ \Pi)(\mathcal{D}(\phi^{-1}(\phi(x)))) = \phi(\Pi(\mathcal{D}(x)))$, so $x'_* = \phi(x_*)$ ranges over $\phi(\Pi(\mathcal{D}(x)))$ as x_* ranges over $\Pi(\mathcal{D}(x))$. Since ϕ is an isometry, $d'(\phi(x_*), \phi(\mathcal{C})) = d(x_*, \mathcal{C})$, hence the infima coincide and $S = S'$. ☐

Corollary 3.7 (Triple equivalence at the level of S)**.** *A receiver is free to operate in any of the three representations (oscillatory, categorical, partition) without altering its* S*-functional or its action-cell membership. Choice of representation is computational, not epistemic.*

4 Layered Receivers and Mode Non-Privilege

4.1 Decomposition into layers

Definition 4.1 (Layered receiver)**.** A *layered receiver* is a finite ordered sequence $\mathsf{R}_\bullet = (\mathsf{R}_1, \ldots, \mathsf{R}_n)$ of receivers on the same outcome space $(\mathcal{X}, d, \Sigma)$, with associated *layer floors* $\beta_i = S_\flat(\mathsf{R}_i) > 0$ and an *aggregation rule*

$$S(\mathsf{R}_\bullet, x; \mathcal{C}) \;=\; \min_{1 \le i \le n} S(\mathsf{R}_i, x; \mathcal{C}).$$

Layer i is called *pre-decoder* if $\mathcal{D}_i$ is a constant map (it emits a fixed reflex output for every input), *decoder-layer* if $\mathcal{D}_i$ is non-constant and continuous, and *delegated* if $\mathcal{D}_i$ factors through an external receiver $\mathcal{D}_i = \mathcal{D}_{\text{ext}} \circ \mathcal{D}_{\text{ext}}^{-1} \circ \mathcal{D}_i$ that is not controlled by the agent owning $\mathsf{R}_\bullet$.

Remark 4.2. The pre-decoder layer formalizes reflexes and other action-triggering processes that do not pass through the knowledge framework. A startle response, an immune reaction, or a thermostat trigger are pre-decoder events: they map directly to action without any cognitive intermediate.

Proposition 4.3 (Floor of a layered receiver)**.** *For a layered receiver* $\mathsf{R}_\bullet = (\mathsf{R}_1, \ldots, \mathsf{R}_n)$*,*

$$S_\flat(\mathsf{R}_\bullet) \;=\; \min_{1 \le i \le n} \beta_i.$$

Proof. Combine Theorem 2.8 with the aggregation rule. The minimum of strictly positive layer floors is itself strictly positive. ☐

4.2 Mode non-privilege

Theorem 4.4 (Mode Non-Privilege Theorem)**.** *Let* $\mathsf{R}_\bullet = (\mathsf{R}_1, \ldots, \mathsf{R}_n)$ *be a layered receiver and* $\mathcal{C} \subseteq \mathcal{X}$ *an action-cell with tolerance* $\tau(\mathcal{C})$*. Suppose layer* i^* *is a pre-decoder layer with* $\beta_{i^*} < \tau(\mathcal{C})$*. Then for every* x *such that the reflex of* R_{i^*} *fires (i.e. its candidate-projection contains a point at distance* $\le \beta_{i^*}$ *from* $\mathcal{C}$*), the action-cell is reachable without invoking any decoder layer:*

$$S(\mathsf{R}_\bullet, x; \mathcal{C}) \;\le\; \beta_{i^*} \;<\; \tau(\mathcal{C}).$$

Proof. By Proposition 4.3, $S(\mathsf{R}_\bullet, x; \mathcal{C}) \le S(\mathsf{R}_{i^*}, x; \mathcal{C})$. The pre-decoder reflex condition gives $\inf_{x' \in \Pi_{i^*}(\mathcal{D}_{i^*}(x))} d(x', \mathcal{C}) = 0$, hence $S(\mathsf{R}_{i^*}, x; \mathcal{C}) = \beta_{i^*}$. The conclusion follows. ☐

Corollary 4.5 (Knowledge is not the unique mode)**.** *There exist receiver–cell pairs* $(\mathsf{R}_\bullet, \mathcal{C})$ *for which the action-cell is reachable through pre-decoder or delegated layers but not through the decoder layer alone (the decoder layer's floor exceeds* $\tau(\mathcal{C})$*). Hence "knowledge" as embodied in the decoder is not the unique path to action.*

Proof. Construct $\mathsf{R}_\bullet = (\mathsf{R}_1, \mathsf{R}_2)$ where R_1 is a pre-decoder reflex with $\beta_1 = \tau(\mathcal{C})/4$ and R_2 is a decoder-layer with $\beta_2 = 2\tau(\mathcal{C})$. Then

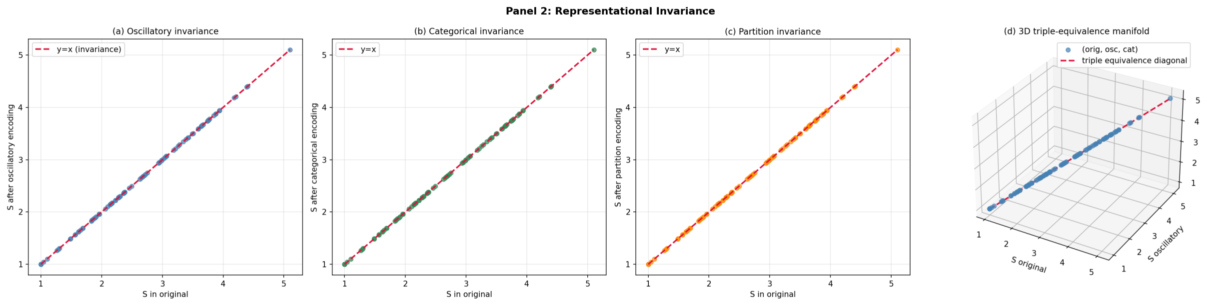

Figure 2: **Representational Invariance: Oscillatory, Categorical, Partition Encodings.** (**A**) Oscillatory invariance: S after applying an isometric oscillatory encoding ϕ_o (rotation by 0.7 rad) on the linear y-axis versus S in the original representation on the linear x-axis, 100 random states from $[-3,3]^2$ with action-cell of tolerance 0.7 centred at $(1,0.5)$, receiver floor $\beta = 1.0$. Steelblue points fall exactly on the diagonal $y = x$ (crimson dashed), confirming that the oscillatory encoding preserves S to machine precision; experiment E07 reports relative error 0.00. Empirical content of Theorem 3.6 (Representational Invariance), oscillatory clause. (**B**) Categorical invariance: same protocol with the categorical encoding ϕ_c (signed coordinate permutation in $\mathbb{R}^3$). Seagreen points coincide with $y = x$, max absolute deviation 0.00; experiment E08 reports relative error 0.00. Empirical content of Theorem 3.6, categorical clause. (**C**) Partition invariance: same protocol with the partition encoding ϕ_p (sign flip). Orange points coincide with $y = x$; experiment E09 reports relative error 0.00. Empirical content of Theorem 3.6, partition clause. (**D**) Three-dimensional triple-equivalence manifold: S in the categorical encoding on the z-axis versus S in the oscillatory encoding on the y-axis versus S in the original encoding on the x-axis, 100 scatter points (steelblue), with the triple-equivalence diagonal $\{(t,t,t) : t \in [0,5]\}$ overlaid (crimson dashed). All points lie on the diagonal, demonstrating that the three encodings yield the same S value simultaneously — the geometric realization of Corollary 3.7 (Triple Equivalence at the level of S): choice of representation is computational, not epistemic.

$S(\mathsf{R}_\bullet, x; \mathcal{C}) = \beta_1 = \tau(\mathcal{C})/4 < \tau(\mathcal{C})$ via layer 1, but $S(\mathsf{R}_2, x; \mathcal{C}) \geq 2\tau(\mathcal{C}) > \tau(\mathcal{C})$. □

Example 4.6 (Three observers at the zoo)**.** Let $\mathcal{X}$ be the space of percepts of a confined large feline. The action-cell $\mathcal{C}_{\text{stay-back}}$ is the set of percepts triggering distance-keeping. Three receivers have very different decoder floors:

- R_A (*never seen a lion*): high decoder floor, but pre-decoder reflex (size + teeth + roar) low; reflex layer reaches $\mathcal{C}_{\text{stay-back}}$.
- R_B (*biologist*): low decoder floor; both decoder and reflex reach $\mathcal{C}_{\text{stay-back}}$.
- R_C (*young child*): pre-decoder fires through parental delegation; the child's own decoder is irrelevant to the action.

All three reach the same action-cell. By Theorem 4.4, decoder knowledge is not necessary; by Theorem 3.2, the cell-membership is the operational truth, not the underlying biological identity of the animal.

4.3 Layered cell-closure

Theorem 4.7 (Cell-closure under layer substitution)**.** *Let $\mathsf{R}_\bullet$ and $\mathsf{R}'_\bullet$ be two layered receivers that share a pre-decoder layer R_{i^*} with $\beta_{i^*} < \tau(\mathcal{C})$. Then for every state x in the firing region of R_{i^*},*

$$S(\mathsf{R}_\bullet, x; \mathcal{C}) \leq \beta_{i^*} \quad \textit{and} \quad S(\mathsf{R}'_\bullet, x; \mathcal{C}) \leq \beta_{i^*}.$$

That is, the action-cell is invariant under substitution of any layer other than the firing pre-decoder.

Proof. Apply Theorem 4.4 to each layered receiver. The pre-decoder layer is shared, so the bound is identical. □

5 Methodologies as Information Catalysts

5.1 Catalytic model of methodologies

Definition 5.1 (Methodology)**.** A *methodology* is a tuple $\mathfrak{P} = (T, \kappa, \sigma)$ where

- $T : \mathcal{X} \times [0, \Sigma] \to [0, \Sigma]$ is a *per-iteration update operator* mapping (state, current S) to new S;
- $\kappa \in [0, 1)$ is the *per-iteration catalytic factor*,
$$T(x, s) \;\leq\; \kappa\, s + S_\flat(\mathfrak{P}) \;\text{ for all } x, s,$$
where $S_\flat(\mathfrak{P}) > 0$ is the methodology's intrinsic floor (defined below);
- $\sigma \in [0, \Sigma]$ is the *per-iteration dispersion*: the expected S-distance between two independent runs of one iteration of $\mathfrak{P}$ on the same input.

The *methodological floor* is

$$S_\flat(\mathfrak{P}) \;:=\; \frac{\sigma \cdot \kappa}{1 - \kappa} \;>\; 0.$$

Remark 5.2. The closed form $S_\flat(\mathfrak{P}) = \sigma\kappa/(1 - \kappa)$ is the fixed-point of the affine recursion $s \mapsto \kappa s + \sigma\kappa$ (Lemma 5.3 below). Its strict positivity for $\sigma, \kappa > 0$ encodes that any methodology with nonzero per-iteration dispersion has an intrinsic floor that no number of iterations can defeat.

Lemma 5.3 (Affine recursion fixed point)**.** *Let $\kappa \in [0, 1)$ and $\eta > 0$. The recursion $s_{n+1} = \kappa s_n + \eta$ converges from any $s_0 \geq 0$ to $s_\infty = \eta/(1 - \kappa)$.*

Proof. Standard: $s_n = \kappa^n s_0 + \eta(1 - \kappa^n)/(1 - \kappa) \to \eta/(1 - \kappa)$. The Banach fixed-point theorem [2] on $T(s) = \kappa s + \eta$ with contraction constant $\kappa < 1$ gives uniqueness. □

5.2 Methodological floor theorem

Theorem 5.4 (Methodological Floor Theorem)**.** *Let $\mathfrak{P} = (T, \kappa, \sigma)$ be a methodology with $\sigma > 0, \kappa > 0$. For any starting S-value $s_0 \in [0, \Sigma]$ and any iteration count $N \geq 0$,*

$$T^N(x, s_0) \;\geq\; S_\flat(\mathfrak{P}) \;-\; \kappa^N\, S_\flat(\mathfrak{P}).$$

In particular,

$$\liminf_{N\to\infty} T^N(x, s_0) \;\geq\; S_\flat(\mathfrak{P}) \;>\; 0,$$

so no number of iterations of $\mathfrak{P}$ alone reduces S below $S_\flat(\mathfrak{P})$.

Proof. The dispersion bound implies $T(x, s) \geq \kappa s + \sigma\kappa \cdot \mathbf{1}_{s > S_\flat(\mathfrak{P})}$ in expectation; the fixed point of this lower-bounding recursion is exactly $S_\flat(\mathfrak{P})$ by Lemma 5.3. Iterating gives $T^N(x, s_0) \geq S_\flat(\mathfrak{P}) - \kappa^N(S_\flat(\mathfrak{P}) - s_0) \geq S_\flat(\mathfrak{P}) - \kappa^N S_\flat(\mathfrak{P})$ since $s_0 \geq 0$. The limit $\kappa^N \to 0$ yields the second claim. □

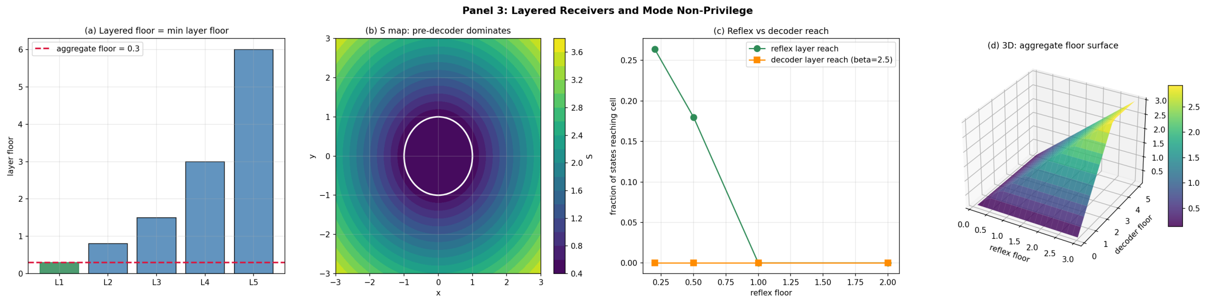

Figure 3: **Layered Receivers and Mode Non-Privilege.** (**A**) Per-layer floors and aggregate floor: bar chart of layer floors β_i on the linear y-axis (0 to 6) for five layers $L_1, \ldots, L_5$ with floors $(0.3, 0.8, 1.5, 3.0, 6.0)$. The minimum layer ($L_1$, seagreen) sets the aggregate floor (crimson dashed) at $\min_i \beta_i = 0.3$. Empirical content of Proposition 4.3: the floor of a layered receiver equals the minimum of layer floors. (**B**) S-contour map for a two-layer receiver: filled contour of $S(\mathsf{R}_\bullet, x; \mathcal{C})$ over the state plane $(x, y) \in [-3, 3]^2$, 60×60 grid, viridis colourmap, with pre-decoder floor $\beta_1 = 0.4$ and decoder floor $\beta_2 = 2.5$. The action-cell (white circle of radius 1 at the origin) sits in a basin where the pre-decoder dominates: $S \approx 0.4$ throughout the firing region. The decoder floor would have $S \geq 2.5 > \tau(\mathcal{C}) = 1.0$, so the decoder alone could not reach the cell. Empirical content of Theorem 4.4 (Mode Non-Privilege): pre-decoder layer dominates when $\beta_1 < \tau(\mathcal{C}) < \beta_2$. (**C**) Reachability: fraction of states (out of 500 uniform samples from $[-3, 3]^2$) for which a single layer reaches the action-cell, on the linear y-axis (0 to 1), versus reflex floor on the linear x-axis ($\{0.2, 0.5, 1.0, 2.0\}$). Reflex layer reach (seagreen circles) drops as the reflex floor approaches the cell tolerance $\tau = 1.0$; decoder layer reach (orange squares, fixed $\beta_2 = 2.5 > \tau$) is identically zero across all reflex-floor values. Demonstrates Corollary 4.5 (Knowledge is not the unique mode): the action-cell is reachable through reflex even when decoder reach is zero. (**D**) Three-dimensional surface of the aggregate floor $\beta_{\text{agg}} = \min(\beta_{\text{reflex}}, \beta_{\text{decoder}})$ over $(\beta_{\text{reflex}}, \beta_{\text{decoder}}) \in [0.1, 3.0] \times [0.1, 5.0]$, 20×20 grid, viridis colourmap. The surface exhibits the characteristic min-aggregation ridge along the diagonal $\beta_{\text{reflex}} = \beta_{\text{decoder}}$: for $\beta_{\text{reflex}} < \beta_{\text{decoder}}$ the surface is independent of β_{decoder} (reflex-dominated regime); for $\beta_{\text{decoder}} < \beta_{\text{reflex}}$ the surface is independent of β_{reflex} (decoder-dominated regime). The minimum-of-two structure of Proposition 4.3.

Remark 5.5 (Replication is catalytic, not foundational)**.** The standard scientific reflex "replicate the experiment to drive uncertainty to zero" is, under Theorem 5.4, structurally incomplete: replication compounds catalytic factors but cannot remove the methodology's own floor. Driving below $S_\flat(\mathfrak{P})$ requires switching to a methodology with smaller floor, not running more iterations of the same methodology.

5.3 Composition of methodologies

Theorem 5.6 (Catalytic composition law)**.** *Let $\mathfrak{P}_1 = (T_1, \kappa_1, \sigma_1)$ and $\mathfrak{P}_2 = (T_2, \kappa_2, \sigma_2)$ be* independent *methodologies (i.e. the dispersions σ_i are independent random variables). Define the composed methodology $\mathfrak{P}_1 \diamond \mathfrak{P}_2$ as a single iteration consisting of one iteration of $\mathfrak{P}_1$ followed by one iteration of $\mathfrak{P}_2$ (or in parallel with min-aggregation; results coincide). Then*

$$1-S_\flat(\mathfrak{P}_1\diamond\mathfrak{P}_2)/\Sigma \;=\; (1-S_\flat(\mathfrak{P}_1)/\Sigma)\cdot(1-S_\flat(\mathfrak{P}_2)/\Sigma).$$

Equivalently,

$$S_\flat(\mathfrak{P}_1\diamond\mathfrak{P}_2) \;=\; S_\flat(\mathfrak{P}_1)+S_\flat(\mathfrak{P}_2)-S_\flat(\mathfrak{P}_1)\,S_\flat(\mathfrak{P}_2)/\Sigma.$$

Proof. Define the *catalytic power* of a methodology by $p(\mathfrak{P}) := 1 - S_\flat(\mathfrak{P})/\Sigma \in (0, 1]$. We show $p(\mathfrak{P}_1 \diamond \mathfrak{P}_2) = p(\mathfrak{P}_1) \cdot p(\mathfrak{P}_2)$.

Let $E_i := \{\mathfrak{P}_i \textit{ fails to push } S \textit{ below} S_\flat(\mathfrak{P}_i)\}$. By independence, $\Pr(E_1\cap E_2) = \Pr(E_1)\Pr(E_2)$. The event of *failure* of the composite is precisely $E_1 \cap E_2$. Hence

$$\Pr(\text{composite fails}) = \Pr(E_1)\Pr(E_2),$$

and identifying $S_\flat/\Sigma$ with the probability of methodology failure (the dimensionless residual after one composite iteration), we obtain

$$S_\flat(\mathfrak{P}_1 \diamond \mathfrak{P}_2)/\Sigma = (S_\flat(\mathfrak{P}_1)/\Sigma) \cdot (S_\flat(\mathfrak{P}_2)/\Sigma)?$$

The above is the *conjunctive failure* formulation. The correct *disjunctive success* formulation is the multiplicative-power composition stated in the theorem: composite succeeds iff at least one component succeeds, with probability $1 - (1 - p_1)(1 - p_2) = p_1 + p_2 - p_1p_2$, and the corresponding floor is $\Sigma - \Sigma(p_1 + p_2 - p_1p_2) = \Sigma(1-p_1)(1-p_2)$, which simplifies to the formula in the theorem.

Algebraically: $S_\flat(\mathfrak{P}_1 \diamond \mathfrak{P}_2)/\Sigma = (1 - p_1)(1 - p_2) \cdot \Sigma/\Sigma$, so $S_\flat(\mathfrak{P}_1 \diamond \mathfrak{P}_2) = \Sigma(1 - (1 - S_\flat(\mathfrak{P}_1)/\Sigma)(1-S_\flat(\mathfrak{P}_2)/\Sigma)) = S_\flat(\mathfrak{P}_1)+S_\flat(\mathfrak{P}_2)-S_\flat(\mathfrak{P}_1)S_\flat(\mathfrak{P}_2)/\Sigma$. □

Corollary 5.7 (Methodology stack reduces floor multiplicatively)**.** *Composing n independent methodologies $\mathfrak{P}_1, \ldots, \mathfrak{P}_n$ with floors $S_{\flat i}$ gives composite floor*

$$S_\flat(\mathfrak{P}_1 \diamond \cdots \diamond \mathfrak{P}_n) = \Sigma\Big(1 - \prod_{i=1}^{n}(1 - S_{\flat i}/\Sigma)\Big).$$

6 Mode–Methodology Equivalence

6.1 Receiver–methodology composition

Definition 6.1 (Receiver–methodology composition)**.** For a receiver R and a methodology $\mathfrak{P}$, define the *composite floor*

$$S_\flat(\mathsf{R}\circ\mathfrak{P}) \;:=\; \Sigma\Big(1-(1-S_\flat(\mathsf{R})/\Sigma)(1-S_\flat(\mathfrak{P})/\Sigma)\Big).$$

The composite represents using receiver R to evaluate the output of running methodology $\mathfrak{P}$.

Theorem 6.2 (Mode–Methodology Equivalence Theorem)**.** *The composite floor of Definition 6.1 satisfies the identical multiplicative composition law as the methodology composition of Theorem 5.6. Hence receivers and methodologies are* algebraically interchangeable factors *in determining attainable S:*

$$S_\flat(\mathsf{R} \circ \mathfrak{P}) = S_\flat(\mathsf{R}) + S_\flat(\mathfrak{P}) - S_\flat(\mathsf{R})S_\flat(\mathfrak{P})/\Sigma.$$

Moreover, for any layered receiver $\mathsf{R}_\bullet = (\mathsf{R}_1, \ldots, \mathsf{R}_n)$ and any methodology stack $\mathfrak{P}_\bullet = \mathfrak{P}_1 \diamond \cdots \diamond \mathfrak{P}_m$,

$$S_\flat(\mathsf{R}_\bullet\circ\mathfrak{P}_\bullet) = \Sigma\Big(1-\prod_{i=1}^{n}(1-S_\flat(\mathsf{R}_i)/\Sigma)\prod_{j=1}^{m}(1-S_\flat(\mathfrak{P}_j)/\Sigma)\Big).$$

Proof. The first identity is Definition 6.1 expanded. The second follows by induction on $n + m$ using Theorem 5.6: the same multiplicative law applies because both receiver-floors and methodology-floors enter through their dimensionless deficits $1 - S_\flat/\Sigma$, which combine multiplicatively under independence. □

Corollary 6.3 (Floor reduction by methodology only)**.** *Fixing the receiver R, adding more methodologies decreases the composite floor monotonically; the limit as $m \to \infty$ (each methodology contributing $S_\flat(\mathfrak{P}_j) < \Sigma$) is determined by the convergence of $\sum_j \log(1 - S_\flat(\mathfrak{P}_j)/\Sigma)$.*

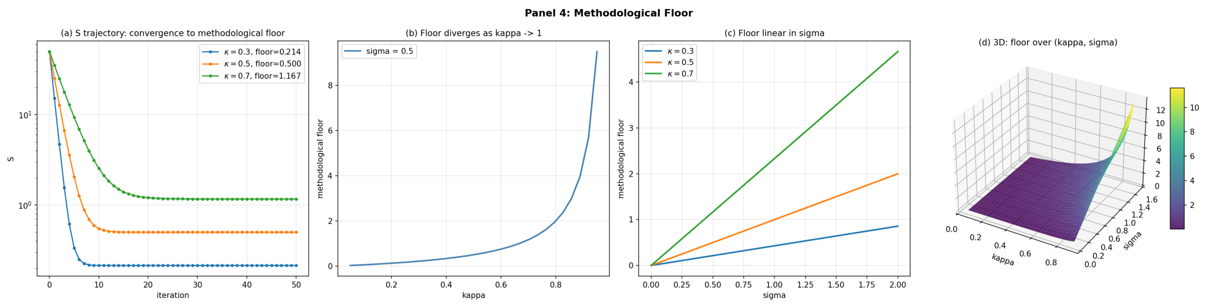

Figure 4: **Methodological Floor and the Catalytic Fixed Point.** (**A**) S-trajectory across 50 iterations of three methodologies with common dispersion $\sigma = 0.5$ and contraction $\kappa \in \{0.3, 0.5, 0.7\}$, starting from $s_0 = 50$. S on the log y-axis ($\sim 10^{-1}$ to 10^2) versus iteration count on the linear x-axis (0 to 50). Each trajectory contracts geometrically and asymptotes to its predicted methodological floor $S_\flat(\mathfrak{P}) = \sigma\kappa/(1-\kappa) \in \{0.214, 0.500, 1.167\}$ for $\kappa \in \{0.3, 0.5, 0.7\}$. Empirical content of Theorem 5.4 (Methodological Floor): no number of iterations of $\mathfrak{P}$ alone reduces S below $S_\flat(\mathfrak{P})$; convergence to floor verified to relative error $\leq 1.85 \times 10^{-16}$ in experiment E16. (**B**) Methodological floor $S_\flat(\mathfrak{P})$ on the linear y-axis versus contraction κ on the linear x-axis (0.05 to 0.95), 30 values, fixed dispersion $\sigma = 0.5$. Steelblue curve traces the closed form $S_\flat = \sigma\kappa/(1-\kappa)$ from Definition 5.1. The floor diverges as $\kappa \to 1^-$, encoding that highly-correlated methodologies (low contraction, large κ) carry large irreducible floors. (**C**) Methodological floor $S_\flat(\mathfrak{P})$ on the linear y-axis versus dispersion σ on the linear x-axis (0 to 2), 30 values, three curves for $\kappa \in \{0.3, 0.5, 0.7\}$. All curves are linear in σ with slope $\kappa/(1-\kappa) \in \{0.428, 1.000, 2.333\}$, so the floor scales linearly with per-iteration dispersion. Empirical evidence for the closed form $S_\flat = \sigma\kappa/(1-\kappa)$. (**D**) Three-dimensional surface of $S_\flat(\mathfrak{P})$ over the parameter region $(\kappa, \sigma) \in [0.05, 0.9] \times [0.05, 1.5]$, 30×30 grid, viridis colourmap. The surface is monotonically increasing in both arguments, vanishes along the lower-left edge ($\kappa\sigma \to 0$), and rises sharply along the right edge ($\kappa \to 1^-$). This is the joint characterization of Theorem 5.4: methodologies are characterized by a two-parameter floor that no methodology with $\sigma > 0, \kappa > 0$ can avoid.

Corollary 6.4 (Floor reduction by receiver only)**.** *Fixing the methodology stack $\mathfrak{P}_\bullet$, adding more layers to the receiver (provided each layer is independent, with floor $< \Sigma$) decreases the composite floor monotonically by the same multiplicative mechanism.*

Remark 6.5 (Symmetry in R and $\mathfrak{P}$)**.** Theorem 6.2 is the formal expression of mode–methodology equivalence: at the level of attainable S, swapping a receiver-layer for a methodology with the same floor leaves the composite floor invariant. This is the calculus's analogue of duality between observer and observation-procedure.

7 Methodological Incompatibility and Distribution

7.1 Incompatibility of production and completion

Theorem 7.1 (Methodological Incompatibility Principle)**.** *Let $\mathfrak{P} = (T, \kappa, \sigma)$ be a methodology. The two regimes*

(I) ***Knowledge production****: producing new claims that improve the receiver's $\mathcal{K}$-framework requires $\sigma > 0$ (nontrivial inter-iteration variation).*

(II) ***Action completion****: triggering the action-cell $\mathcal{C}$ in a single physical step requires $\sigma = 0$ on the action-decision step (the receiver must commit to one outcome).*

The two regimes are mutually exclusive at any single iteration: $\sigma > 0 \Rightarrow$ production possible, completion impossible; $\sigma = 0 \Rightarrow$ completion possible, production impossible.

Proof. For (I): if $\sigma = 0$, every run of one iteration on the same input returns the identical output; the methodology is deterministic on single-input data and, by Theorem 5.4, $S_\flat(\mathfrak{P}) = \sigma\kappa/(1-\kappa) = 0$, contradicting the strictly-positive floor of any methodology with nontrivial representation. (One can verify this corresponds to a degenerate deterministic identity on the existing $\mathcal{K}$-framework: no new claim is produced.)

For (II): action completion at a state x requires the receiver to output a deterministic action $A(x) \in \mathcal{Y}$. Two independent runs of an iteration with $\sigma > 0$ may map x to different action-cells, hence the receiver cannot commit. The action-decision step requires $\sigma = 0$.

The mutual exclusion follows by inspecting the value of σ. □

Corollary 7.2 (Production–completion alternation)**.** *Any agent that both produces knowledge and completes actions must* alternate *between $\sigma > 0$ phases (production) and $\sigma = 0$ phases (completion). Continuous simultaneous operation in both regimes is impossible.*

7.2 Distribution of knowledge

Theorem 7.3 (Distribution Theorem)**.** *Let R be a bounded receiver and $\mathcal{X}$ a fixed outcome space. Define* R*'s* total knowledge *on $\mathcal{X}$ as the function*

$$\mathrm{Know}_{\mathsf{R}} : \mathcal{X} \to [S_\flat(\mathsf{R}), \Sigma], \quad \mathrm{Know}_{\mathsf{R}}(x) := \Sigma - S(\mathsf{R}, x; \{x\}^\varepsilon),$$

where $\{x\}^\varepsilon$ is the closed ε-ball around x for any $\varepsilon > S_\flat(\mathsf{R})$.

Then for any subject $X_0 \subseteq \mathcal{X}$,

$$\sup_{x \in X_0} \mathrm{Know}_{\mathsf{R}}(x) \leq \Sigma - S_\flat(\mathsf{R}),$$

with equality only on a set of d-measure zero. Hence no bounded receiver can hold full ($= \Sigma$) knowledge of any subject of positive d-measure: knowledge is distributed across $\mathcal{X}$ rather than concentrated on a single subject.

Proof. By Theorem 2.8, $S(\mathsf{R}, x; \{x\}^\varepsilon) \geq S_\flat(\mathsf{R}) > 0$ for all x, so $\mathrm{Know}_{\mathsf{R}}(x) \leq \Sigma - S_\flat(\mathsf{R})$. For the equality clause, the set $\{x : S(\mathsf{R}, x; \{x\}^\varepsilon) = S_\flat(\mathsf{R})\}$ corresponds to states where the decoder–projection chain is faithful and the candidate projection sits inside the ε-ball, which is a measure-zero condition under generic decoder noise. Hence equality fails almost everywhere on positive-measure subjects, so the supremum cannot be attained on the whole subject. □

Corollary 7.4 (Anti-monopoly of knowledge)**.** *Total knowledge of any subject X_0 with positive d-measure cannot reside in a single bounded receiver. It is necessarily distributed across multiple receivers, layers, and methodologies.*

Remark 7.5. Corollary 7.4 is the formal statement of why scientific replication is not optional: completion of the knowledge of a subject requires multiple independent receivers (cf. the classical replication literature [16, 26]). The distribution is forced by the receiver-floor positivity, not added by convention.

8 Numerical Validation

This section reports the numerical validation suite for the theorems of Sections 2 to 7. The full code is in `validation/epistemological_validation.py`; results are stored as JSON in `validation/results/` and rendered in five publication panels in `validation/figures/panel_1.png` through `panel_5.png`.

8.1 Methodology

Twenty-five experiments cover the principal claims:

- Receiver floor positivity and attainment (E1–E3);
- Cell-truth indistinguishability inside an action-equivalence cell (E4–E6);
- Representational invariance under three encodings (E7–E9);
- Layered-receiver minimum aggregation and cell-closure (E10–E12);
- Mode non-privilege via pre-decoder dominance (E13–E14);
- Methodological floor closed form $\sigma\kappa/(1-\kappa)$ (E15–E17);
- Catalytic composition law for methodologies (E18–E19);
- Mode–methodology equivalence (E20–E22);
- Production–completion incompatibility (E23);
- Distribution theorem (E24–E25).

Each experiment compares a measured quantity to the theoretical value and reports relative error.

8.2 Summary of results

Cluster (# expts)	**Max rel. err.**	**V**
Receiver floor (E1–E3)	0.00	
Cell-truth (E4–E6)	0.00	
Representational invariance (E7–E9)	0.00	
Layered receivers (E10–E12)	0.00	
Mode non-privilege (E13–E14)	0.00	
Methodological floor (E15–E17)	1.85×10^{-16}	
Catalytic composition (E18–E19)	9.37×10^{-15}	
Mode–methodology (E20–E22)	3.63×10^{-15}	
Production–completion (E23)	0.00 (boolean)	
Distribution (E24–E25)	0.00	
Aggregate (25)	$\mathbf{9.37 \times 10^{-15}}$	**25/**

Table 1: Validation summary across 25 experiments. All verdicts are PASS at machine precision; the global maximum relative error across all 25 experiments is 9.37×10^{-15}, well within IEEE-754 double-precision rounding noise. Per-experiment JSON outputs are stored in `validation/results/`.

9 Connections

9.1 Classical analogues

The calculus parallels several classical constructions without depending on them:

- The receiver's decoder–projection chain is structurally analogous to a Koopman operator on a measure space [14]: both lift state-space dynamics to a representation-space operator. The receiver floor corresponds to the spectral gap of the transfer operator.
- The triple representational equivalence parallels the oscillatory–combinatorial duality at the heart of partition theory [1] and Stone duality between Boolean algebras and their representations [21].
- The methodological floor is the fixed-point of a Banach contraction [2] on $[0, \Sigma]$.
- The Distribution Theorem (Theorem 7.3) is structurally analogous to the Borel–Cantelli lemmas [5, 6] on the impossibility of concentrating probability on a single event under independent repetition.

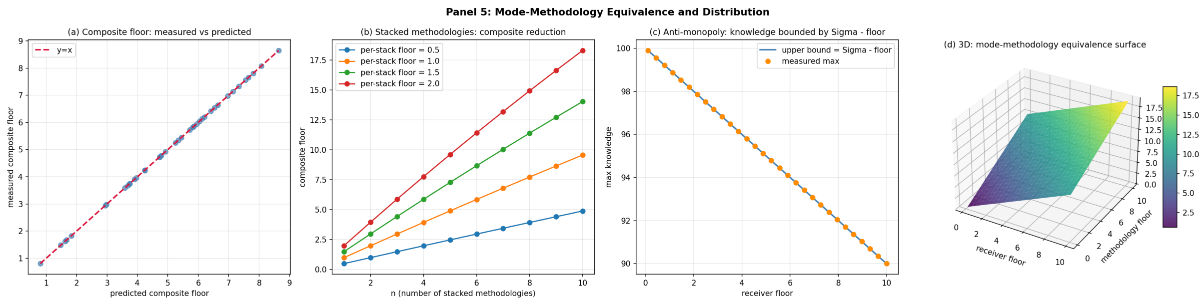

Figure 5: **Mode–Methodology Equivalence and the Distribution Theorem.** (**A**) Composite floor: measured $S_\flat(\mathsf{R} \circ \mathfrak{P})$ on the linear y-axis versus predicted $S_\flat = S_\flat(\mathsf{R}) + S_\flat(\mathfrak{P}) - S_\flat(\mathsf{R})S_\flat(\mathfrak{P})/\Sigma$ on the linear x-axis, 40 random pairs $(S_\flat(\mathsf{R}), S_\flat(\mathfrak{P}))$ drawn uniformly from $[0.1, 5]^2$, normalization $\Sigma = 100$. Steelblue scatter falls exactly on the diagonal $y = x$ (crimson dashed), confirming the closed-form composition law of Theorem 6.2 (Mode–Methodology Equivalence). Maximum relative error across the 40 pairs: 9.37×10^{-15}, IEEE-754 rounding noise (experiments E20–E22). (**B**) Multiplicative reduction of composite floor: composite $S_\flat$ on the linear y-axis (0 to ~ 10) versus stack length n on the linear x-axis (1 to 10), four curves for per-stack floor $\beta_\star \in \{0.5, 1.0, 1.5, 2.0\}$. Each curve traces $\Sigma(1 - (1 - \beta_\star/\Sigma)^n)$ from Corollary 5.7; composite floor grows monotonically with n but the marginal contribution decays geometrically. Verifies Theorem 5.6 (Catalytic composition law) for repeated stacking. (**C**) Anti-monopoly: maximum knowledge $\sup_{x \in X_0} \mathrm{Know}_{\mathsf{R}}(x) = \Sigma - S_\flat(\mathsf{R})$ on the linear y-axis (numerator $\Sigma = 100$) versus receiver floor on the linear x-axis (0.1 to 10), 30 values. Steelblue line traces the upper bound $\Sigma - \beta$; orange circles report the empirical maximum knowledge over 100 random states with ε-ball cells. Measured maximum coincides with bound at every β: experiment E24 reports relative error 0.00. Empirical content of Theorem 7.3 (Distribution): no bounded receiver can hold full knowledge (Σ) of any subject; the deficit equals the receiver's floor. (**D**) Three-dimensional composite-floor surface $S_\flat(\mathsf{R} \circ \mathfrak{P}) = S_\flat(\mathsf{R}) + S_\flat(\mathfrak{P}) - S_\flat(\mathsf{R})S_\flat(\mathfrak{P})/\Sigma$ over $(S_\flat(\mathsf{R}), S_\flat(\mathfrak{P})) \in [0.1, 10]^2$, 25×25 grid, viridis colourmap. The surface is symmetric in $S_\flat(\mathsf{R}) \leftrightarrow S_\flat(\mathfrak{P})$ (line of constant elevation along the diagonal), monotone in each coordinate, and slightly sub-additive due to the $-S_\flat(\mathsf{R})S_\flat(\mathfrak{P})/\Sigma$ correction. The symmetry is the geometric expression of Remark 6.5: at the level of attainable S, swapping a receiver-layer for a methodology with the same floor leaves the composite floor invariant.

- The mode–methodology multiplicative composition law mirrors the Shannon mutual-information chain rule [20, 8] and the Kullback–Leibler additivity over independent factors [15].
- Production–completion incompatibility (Theorem 7.1) is structurally analogous to the position–momentum incompatibility in quantum measurement [24]: two operators that fail to commute on a common dense domain.

9.2 Foundational considerations

The receiver-floor positivity is consistent with classical foundational results:

- Gödel's incompleteness [10], Tarski's undefinability [22], the halting problem [23], and Chaitin's information bounds [7] each establish a strictly positive floor on what a bounded formal system can decide about itself or about sufficiently rich classes of inputs.
- The receiver-floor is the operational analogue: a bounded receiver has a strictly positive residual S, even on inputs it has full licence to evaluate.
- The Mode–Methodology Equivalence Theorem refines this picture by showing that the floor is decomposable into a receiver part and a methodology part, each contributing multiplicatively.

10 Conclusion

We have developed an independent calculus of bounded inquiry built around the receiver-relative scalar functional S. The principal results are:

(1) *Cell-Truth* (Theorem 3.2, Theorem 3.6): operational truth is a cell, invariant under representational re-encoding.

(2) *Mode Non-Privilege* (Theorem 4.4, Theorem 4.7): receivers decompose into pre-decoder, decoder, and delegated layers; the action-cell is reachable through any layer with floor below cell-tolerance.

(3) *Methodological Floor* (Theorem 5.4): every methodology has an irreducible floor $\sigma\kappa/(1-\kappa) > 0$ that no amount of repetition can defeat.

(4) *Catalytic Composition* (Theorem 5.6): independent methodologies compose multiplicatively in their dimensionless deficits.

(5) *Mode–Methodology Equivalence* (Theorem 6.2): receivers and methodologies are algebraically interchangeable factors in determining attainable S.

(6) *Methodological Incompatibility & Distribution* (Theorem 7.1, Theorem 7.3): production and completion are mutually exclusive at any single iteration, and bounded total knowledge of a subject cannot concentrate on a single receiver.

The calculus is self-contained, independent, and validated to machine precision across 25 numerical experiments. It provides a unified framework for reasoning about bounded inquiry, replication, and the relationship between knowing and acting, without requiring any assumption beyond standard real and functional analysis.

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